

ECO 310: Problem 2a(ii) Detailed Explanation

Risk Aversion and Certainty Equivalents

Explanation Notes

February 28, 2026

1 Problem Statement

Problem 2a(ii)

If Mavi is more risk-averse than Melsi, then from Lecture 10, her risk premium is higher for every gamble. **Prove that this implies that for a particular gamble, Melsi's certainty equivalent is higher than Mavi's.**

2 Key Definitions

Before proving the result, let's establish the fundamental concepts:

Definition 1 (Certainty Equivalent (CE)). The certainty equivalent of a gamble is the guaranteed amount of wealth that provides the same utility as the expected utility of the gamble:

$$EU(gamble) = u(CE)$$

Definition 2 (Risk Premium (π)). The risk premium is the maximum amount a decision-maker would pay to avoid the risk and receive the expected value with certainty:

$$EU(gamble) = u(E[W] - \pi)$$

where $E[W]$ is the expected wealth from the gamble.

Definition 3 (Risk Aversion). A person is risk-averse if they prefer a certain outcome to a gamble with the same expected value. More risk-averse individuals have higher risk premiums for any given gamble.

3 The Fundamental Relationship

3.1 Step 1: Connecting CE and π

We have two equations that both equal $EU(gamble)$:

$$EU(gamble) = u(CE) \tag{1}$$

$$EU(gamble) = u(E[W] - \pi) \tag{2}$$

Since both expressions equal $EU(gamble)$, we can equate them:

$$u(CE) = u(E[W] - \pi)$$

3.2 Step 2: Using Monotonicity of Utility

Since utility functions are strictly increasing in wealth (more money is always better), the function $u(\cdot)$ is one-to-one. This means:

$$u(\text{CE}) = u(E[W] - \pi) \implies \text{CE} = E[W] - \pi$$

3.3 Step 3: Rearranging to Isolate CE

From the equation $\text{CE} = E[W] - \pi$, we can rearrange:

$$\boxed{\text{CE} = E[W] - \pi} \tag{3}$$

This is our **key relationship**: the certainty equivalent equals the expected value minus the risk premium.

4 The Proof

Now we can prove the main result.

4.1 Setup

Consider a particular gamble with expected wealth $E[W]$.

Let:

- CE_{Melsi} = Melsi's certainty equivalent for this gamble
- CE_{Mavi} = Mavi's certainty equivalent for this gamble
- π_{Melsi} = Melsi's risk premium for this gamble
- π_{Mavi} = Mavi's risk premium for this gamble

4.2 Given Information

We are told that Mavi is more risk-averse than Melsi, which means:

$$\pi_{\text{Mavi}} > \pi_{\text{Melsi}} \tag{4}$$

4.3 What We Need to Prove

We want to show:

$$\text{CE}_{\text{Melsi}} > \text{CE}_{\text{Mavi}}$$

4.4 Step-by-Step Proof

Proof. Step 1: Apply equation (3) to Melsi:

$$\text{CE}_{\text{Melsi}} = E[W] - \pi_{\text{Melsi}}$$

Step 2: Apply equation (3) to Mavi:

$$\text{CE}_{\text{Mavi}} = E[W] - \pi_{\text{Mavi}}$$

Step 3: Note that both dogs face the same gamble, so the expected wealth $E[W]$ is identical for both.

Step 4: Take the difference between Melsi's and Mavi's certainty equivalents:

$$\begin{aligned} \text{CE}_{\text{Melsi}} - \text{CE}_{\text{Mavi}} &= (E[W] - \pi_{\text{Melsi}}) - (E[W] - \pi_{\text{Mavi}}) \\ &= E[W] - \pi_{\text{Melsi}} - E[W] + \pi_{\text{Mavi}} \\ &= \pi_{\text{Mavi}} - \pi_{\text{Melsi}} \end{aligned}$$

Step 5: From our given information (4), we know $\pi_{\text{Mavi}} > \pi_{\text{Melsi}}$, which means:

$$\pi_{\text{Mavi}} - \pi_{\text{Melsi}} > 0$$

Step 6: Therefore:

$$\text{CE}_{\text{Melsi}} - \text{CE}_{\text{Mavi}} > 0$$

Conclusion:

$\text{CE}_{\text{Melsi}} > \text{CE}_{\text{Mavi}}$

□

5 Intuition

Why does this make sense?

- The **certainty equivalent** is the guaranteed amount you'd accept instead of taking the gamble.
- The **risk premium** is how much you'd pay (or how much you're discounting the expected value) to avoid the risk.
- A more risk-averse person (Mavi) demands a bigger discount for bearing risk, so her risk premium π is higher.
- Since $\text{CE} = E[W] - \pi$, a higher risk premium means a lower certainty equivalent.
- In other words: **Mavi needs more compensation to accept the same level of risk**, so the guaranteed amount that would make her indifferent is lower.
- Conversely, Melsi (less risk-averse) is more willing to accept risk, so she'd be satisfied with a higher certain payment.

6 Connection to Part (i)

This result connects to part (i) of the problem, which showed that someone accepts a bet if and only if $\text{CE} > w_0$ (initial wealth).

Since:

- Both dogs are equally wealthy (same w_0)
- Whenever Mavi accepts a bet, her $\text{CE}_{\text{Mavi}} > w_0$
- Melsi's CE is even higher: $\text{CE}_{\text{Melsi}} > \text{CE}_{\text{Mavi}} > w_0$
- Therefore Melsi also accepts the bet

Conclusion: Any bet that Mavi (more risk-averse) is willing to accept, Melsi (less risk-averse) will also accept.